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Mathematics

On finite subgroups of SU(2), simple Lie algebras, and the McKay correspondence

BERTRAM KOSTANT

Mathematical Sciences Research Institute, Berkeley, CA 94720

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ABSTRACT **The question is considered of how the restriction $\pi_n|\Gamma$, where π_n is irreducible for all n and Γ is a finite subgroup of SU(2), decomposes into Γ-irreducibles for any arbitrary $n \in \mathbb{Z}_+$. It is announced that, using the McKay correspondence, the problem has an elegant solution in terms of the Coxeter element for the associated Lie algebra and that the numbers involved come in a beautiful way from the root structure.**

The classification of all finite subgroups Γ of SU(2) is well known (see, e.g., refs. 1 and 2). Let $S(\mathbb{C}^2) = \oplus_{n=0}^{\infty} S^n(\mathbb{C}^2)$ be the symmetric algebra over $\mathbb{C}^2$ and let π_n be the representation of SU(2) on $S^n(\mathbb{C}^2)$ induced by its action on $\mathbb{C}^2$. One knows that π_n is irreducible for all n and the set of equivalence classes $\{\pi_n\}$, $n = 0, \ldots,$ defines the unitary dual of SU(2). Now let $\Gamma \subseteq$ SU(2) be any finite group. Consider the natural question: How does the restriction $\pi_n|\Gamma$ decompose into Γ-irreducibles for any arbitrary $n \in \mathbb{Z}_+$? This question has arisen recently in connection with the resolution of certain algebraic singularities. Using the well-known theory of complex reflection groups (applicable here by an observation of P. Slodowy), the question may be dealt with by looking separately at the "invariant" and the "harmonic" component of $S(\mathbb{C}^2)$. This has been done by G. Gonzalez-Springberg and J.-L. Verdier in ref. 3 and H. Knörrer in ref. 4 and empirically observed multiplicity tables have been compiled in those papers. However, the interesting numbers involved are unexplained. In this note, using the McKay correspondence, we find that the problem has a very elegant solution in terms of the orbits of the Coxeter element for the associated simple Lie algebra and that the numbers involved come in a beautiful way from the root structure. The results are obtained using Lie theoretic principles and there is no reliance on empirical observations. The proofs will appear elsewhere.

Let $\Gamma \subseteq$ SU(2) be a nontrivial finite subgroup of SU(2) and let $\{\gamma_0, \ldots, \gamma_l\} = \hat{\Gamma}$ be the set of equivalence classes of irreducible finite-dimensional complex representations of Γ. Then if $\gamma:\Gamma \to$ SU(2) is the given two-dimensional representation, one defines an $(l + 1) \times (l + 1)$ matrix $\mathbf{A}(\Gamma)$ with entries in $\mathbb{Z}_+$ by decomposing the tensor product $\gamma_j \otimes \gamma = \Sigma_i \mathbf{A}(\Gamma)_{ij}\gamma_i$ into irreducible components. John McKay has made the remarkable observation (see refs. 5 and 6) that there exists a complex simple Lie algebra $\mathfrak{g} = \mu(\Gamma)$ of rank l such that

$$\mathbf{A}(\Gamma) = 2 - C(\tilde{\mathfrak{g}}),$$

where $\tilde{\mathfrak{g}}$ is the affine Kac–Moody Lie algebra associated to $\mathfrak{g}$ and $C(\tilde{\mathfrak{g}})$ is a Cartan matrix of $\tilde{\mathfrak{g}}$. Moreover the correspondence $\Gamma \to \mu(\Gamma) = \mathfrak{g}$ sets up a bijection between the set of all isomorphism classes of subgroups $\Gamma \subseteq$ SU(2) and the set of all isomorphism classes of complex Lie algebras of type A, D, and E.

The Cartan matrix $C(\tilde{\mathfrak{g}})$ is with respect to a set of simple roots $\alpha_i \in \tilde{\mathfrak{h}}'$, $i = 0, \ldots, l$, where $\mathfrak{h} \subseteq \tilde{\mathfrak{h}}$ are, respectively, Cartan subalgebras of $\mathfrak{g} \subseteq \tilde{\mathfrak{g}}$. The indexing may be chosen so that γ_0 is the trivial representation of Γ and $\alpha_0 \in \tilde{\mathfrak{h}}'$ is the added simple root corresponding to the negative of the highest root $\psi \in \mathfrak{h}'$ of $\mathfrak{g}$. Now if $\pi_n|\Gamma = \Sigma_{i=0}^{l} m_i\gamma_i$ we associate to $\pi_n|\Gamma$ the element $v_n \in \tilde{\mathfrak{h}}'$ in the root lattice defined by putting $v_n = \Sigma_{i=0}^{l} m_i\alpha_i$. The problem we set for ourselves is then the determination of the generating function $P_\Gamma(t)$, with coefficients in $\tilde{\mathfrak{h}}'$, defined by putting

$$P_\Gamma(t) = \sum_{n=0}^{\infty} v_n t^n.$$

We restrict our attention to the cases where the Coxeter number $h = 2g$ of $\mathfrak{g}$ is even (so that $g \in \mathbb{Z}_+$). This excludes only the case where Γ is a cyclic group of odd order. (The latter case is readily dealt with by considering first the cyclic group $\Gamma \times \mathbb{Z}_2$ of even order where $\mathbb{Z}_2 = \{\pm I\}$.) We can now speak of the special node i_* of the Dynkin diagram of $\mathfrak{g}$. If $\mathfrak{g} \not\cong \mathbf{A}_l$ then i_* is the branch point. If $\mathfrak{g} \cong \mathbf{A}_{2m-1}$ ($\mathbf{A}_{2m}$ has been excluded) then i_* is the midpoint. Our main results will be stated in the following sequence of six theorems.

Remark. In the light of Slodowy's observation (see above), the following result (*Theorem 1*) is not new. However, the product decomposition in *Theorem 1* arises not from reflection group theory but from a study of the Coxeter element. As a consequence, the numbers involved are directly related to the root structure of $\mathfrak{g} = \mu(\Gamma)$. But then, if one proceeds to make the connection with the reflection group theory, one obtains as a theorem (not just an observation) that the number of reflecting hyperplanes is given in terms of the Coxeter number and the lesser degree of the two fundamental invariants is given in terms of the highest coefficient of the maximal root of $\mathfrak{g}$.

THEOREM 1. *There exist* $\mathbf{z}_i \in \tilde{\mathfrak{h}}'$, i = 0, …, h *and even integers* $2 \leqslant a \leqslant b \leqslant h$ *such that*

$$P_\Gamma(t) = \frac{\sum_{i=0}^{h} \mathbf{z}_i t^i}{(1 - t^a)(1 - t^b)}.$$

The integers a and b are determined by the next result. Their values are given in Table 1.

Table 1. a and b values for the five types of subgroups of SU(2)

F*	$\mathfrak{g}$	a	b	h
$\mathbf{Z}_n^*$	A_{2n-1}	2	$2n$	$2n$
Δ_n^*	D_{n+2}	4	$2n$	$2n + 2$
A_4^*	E_6	6	8	12
S_4^*	E_7	8	12	18
A_5^*	E_8	12	20	30

If F is a subgroup of SO(3) we denote by F* its inverse image in SU(2) with respect to the usual double covering SU(2) → SO(3). Δ_n, the dihedral group of order $2n$.

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THEOREM 2. *One has* $a = 2d$ *where* $d = d_{i_*}$ *is the coefficient of the highest root* ψ *corresponding to the special node* i_* (1 *for* A_l, 2 *for* D_l, 3 *for* E_6, 4 *for* E_7, *and* 6 *for* E_8). *Furthermore,* b *is given by*

$$b = h + 2 - a.$$

In addition one has

$$ab = 2|\Gamma|.$$

If W is the (finite) Weyl group of $(\mathfrak{h}, \mathfrak{g})$, then one knows that there is a Coxeter element $\sigma \in W$ corresponding to $\Pi = \{\alpha_1, \ldots, \alpha_l\}$ such that $\sigma = \tau_2\tau_1$, where $\tau_2, \tau_1 \in W$ have order at most 2 and correspond to a decomposition $\Pi = \Pi_1 \cup \Pi_2$ into orthogonal sets. The order may be chosen so that $\tau_2\psi = \psi$. For $n \in \mathbb{Z}_+$ let $\tau_n = \tau_1$ if n is odd and $\tau_n = \tau_2$ if n is even. Also let $\tau^{(n)}$ be the alternating decreasing product $\tau^{(n)} = \tau_n\tau_{n-1} \cdots \tau_1$. Let $e = b/2$. The vectors $\mathbf{z}_i$ are determined in

THEOREM 3. *One has* $\mathbf{z}_0 = \mathbf{z}_h = \alpha_0$. *For* $n = 1, \ldots, h-1$ *one has* $\mathbf{z}_n \in \mathfrak{h}'$ (*not just* $\tilde{\mathfrak{h}}'$) *and* $\mathbf{z}_n$ *is given by*

$$\mathbf{z}_n = \tau^{(n-1)}\psi - \tau^{(n)}\psi, \qquad [1]$$

where ψ *is the highest root of* $(\mathfrak{h}, \mathfrak{g})$. *Furthermore,*

$$\mathbf{z}_g = 2\alpha_{i_*}$$

and in general one has the symmetry

$$\mathbf{z}_{g+k} = \mathbf{z}_{g-k}$$

for $k = 1, \ldots, g$. *Finally, if* $n = 1, \ldots, g - 1$, *there exists distinct* $\alpha_{i_1}, \ldots, \alpha_{i_r} \in \Pi_j$, $j \in \{1, 2\}$, *where* $j \equiv n$ *mod* 2, *such that*

$$\mathbf{z}_n = \alpha_{i_1} + \cdots + \alpha_{i_r}.$$

Moreover,

$$r = \begin{cases} 1 \text{ if } 1 \leq n \leq d - 1 \\ 2 \text{ if } d \leq n \leq e - 1 \\ 3 \text{ if } e \leq n \leq g - 1 \end{cases}$$

and in fact $r = 1, 2,$ *or* 3 *according as to whether* $(\tau^{(n)}\psi, \tau^{(n-1)}\psi)$ *is positive, zero, or negative.*

Remark. Instead of generating $\mathbf{z}_n$ by the highest root ψ the $\mathbf{z}_n$ vectors can be generated in a manner similar to Eq. **1** using the simple root α_{i_*} instead of ψ.

Now the Poincaré series $P_\Gamma(t)_i$ for the individual representations γ_i is obviously obtained by considering only the ith coefficient of the vectors v_n. Clearly, by *Theorem 1*, using the subscript i for this coefficient

$$P_\Gamma(t)_i = \frac{\mathbf{z}(t)_i}{(1 - t^a)(1 - t^b)},$$

where $\mathbf{z}(t) = \sum_{n=0}^{h} \mathbf{z}_n t^n$. Thus to, determine $P_\Gamma(t)_i$ we have only to determine $\mathbf{z}(t)_i$. Consider first the case $i = 0$. One notes that $P_\Gamma(t)_0$ is the Poincaré series for the algebra of invariants $S(\mathbb{C}^2)^\Gamma$. The following is known although probably not expressed in terms of the Coxeter number h of $\mathfrak{g}$ and the largest coefficient $d = a/2$ of the highest root ψ of $(\mathfrak{h}, \mathfrak{g})$. It is an immediate consequence of *Theorem 3*.

THEOREM 4. *One has* $\mathbf{z}(t)_0 = 1 + t^h$ *so that*

$$P_\Gamma(t)_0 = \frac{1 + t^h}{(1 - t^a)(1 - t^b)}.$$

Next consider the case where $i = i_*$. One notes that γ_{i_*} is an irreducible representation of maximal dimension of Γ and is the unique such in case $\mu(\Gamma) = E_6, E_7$, or E_8. Observe that the coefficient of t^g is 2 in the following result. This is the only occurrence of a coefficient greater than 1 for any $\mathbf{z}(t)_i$.

THEOREM 5. *One has*

$$\mathbf{z}(t)_{i_*} = \sum_{j=0}^{d-1} t^{g-2j} + \sum_{j=0}^{d-1} t^{g+2j}.$$

In particular, $\mathbf{z}(1)_{i_*} = a$.

Now let Φ be the set of roots φ of $(\mathfrak{h}, \mathfrak{g})$ such that $(\varphi, \psi) > 0$. One has

$$\text{Card } \Phi = 2h - 3$$

and one may refer to Φ as a Heisenberg system since a set of corresponding root vectors span a Heisenberg Lie subalgebra of $\mathfrak{g}$ of dimension $2h - 3$. By intersecting Φ with the l orbits (naturally parametrized by Π) of the Coxeter element σ, one obtains a partition

$$\Phi = \bigcup_{i=1}^{l} \Phi^i.$$

One proves

$$\text{Card } \Phi^i = \begin{cases} 2d_i & \text{if } i \neq i_* \\ 2d_i - 1 & \text{if } i = i_*, \end{cases} \qquad [2]$$

where the d_i are the coefficients of the highest root. That is, $\psi = \sum_{i=1}^{l} d_i\alpha_i$. On the other hand, there is a natural function $\varphi \mapsto n(\varphi) \in \mathbb{Z}_+$ on the set of positive roots defined using the $\tau^{(n)}$. Among other things, it is injective on each Φ^i. The case for the remaining nodes is settled by *Theorem 6*.

THEOREM 6. *If* $i \neq 0$ *or* i_* *then*

$$\mathbf{z}(t)_i = \sum_{\varphi \in \Phi^i} t^{n(\varphi)},$$

so that in particular all the coefficients of $\mathbf{z}(t)_i$ *are either* 1 *or* 0. *Furthermore, the coefficients of* t^{g-k} *and* t^{g+k} *are equal for* $k = 1, \ldots, g$ *and vanish for* $k = 0$. *Finally* (by Eq. **2**), $\mathbf{z}(1)_i = 2d_i$.

The values for the $\mathbf{z}(t)_i$ polynomials obtained after labeling the nodes [eliminating i_* (see *Theorem 5*)] on the Dynkin diagram of E_6, E_7, and E_8 are given in Table 2.

The proofs of the above results will appear in ref. 7. Papers 8–14 are cited in ref. 7.

Table 2. Values for $\mathbf{z}(t)_i$ polynomials

A_4^* (E_6)

1 2 3 4
•—•—•—•
| 5

$\mathbf{z}(t)_1 =$	$t^4 + t^8$
$\mathbf{z}(t)_2 =$	$t^3 + t^5 + t^7 + t^9$
$\mathbf{z}(t)_3 =$	$t^3 + t^5 + t^7 + t^9$
$\mathbf{z}(t)_4 =$	$t^4 + t^8$
$\mathbf{z}(t)_5 =$	$t + t^5 + t^7 + t^{11}$

S_4^* (E_7)

1 2 3 4 5
•—•—•—•—•
| 6

$\mathbf{z}(t)_1 =$	$t^6 + t^{12}$
$\mathbf{z}(t)_2 =$	$t^5 + t^7 + t^{11} + t^{13}$
$\mathbf{z}(t)_3 =$	$t^4 + t^6 + t^8 + t^{10} + t^{12} + t^{14}$
$\mathbf{z}(t)_4 =$	$t^2 + t^6 + t^8 + t^{10} + t^{12} + t^{16}$
$\mathbf{z}(t)_5 =$	$t + t^7 + t^{11} + t^{17}$
$\mathbf{z}(t)_6 =$	$t^4 + t^8 + t^{10} + t^{14}$

A_5^* (E_8)

1 2 3 4 5 6
•—•—•—•—•—•
| 7

$\mathbf{z}(t)_1 =$	$t + t^{11} + t^{19} + t^{29}$
$\mathbf{z}(t)_2 =$	$t^2 + t^{10} + t^{12} + t^{18} + t^{20} + t^{28}$
$\mathbf{z}(t)_3 =$	$t^3 + t^9 + t^{11} + t^{13} + t^{17} + t^{19} + t^{21} + t^{27}$
$\mathbf{z}(t)_4 =$	$t^4 + t^8 + t^{10} + t^{12} + t^{14} + t^{16} + t^{18} + t^{20} + t^{22} + t^{26}$
$\mathbf{z}(t)_5 =$	$t^6 + t^8 + t^{12} + t^{14} + t^{16} + t^{18} + t^{22} + t^{24}$
$\mathbf{z}(t)_6 =$	$t^7 + t^{13} + t^{17} + t^{23}$
$\mathbf{z}(t)_7 =$	$t^6 + t^{10} + t^{14} + t^{16} + t^{20} + t^{24}$

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